

The setup consists of solar arrays with photovoltaic cells placed over the ground. The arrays are connected to a DC pump motor. A storage tank, connected to the pump via pipelines, is installed nearby at a certain height from the ground.

Solar energy captured by PV cells is converted into direct electric current and pushed to the DC pump. A controller built inside the DC pump is used to control and direct the water, be it for drawing from the source (pond, bore well, river, etc) or transferring drawn water to the reservoir tank for conservation. The pump power is usually indicated in Horse Power (HP) or at times watts-peak (Wp).

The pump is chosen based on the type and depth of the water source. Sources from Tata Power Solar say that a 2-acre land can be irrigated using 40,000 litres of water in a day if a 1000Wp pump is installed to draw from a 10-metre deep water source.

The benefits

Electricity supply in India is still unstable in many rural and semi-urban areas. This makes irrigating farmlands with water pumped through grid electricity a far cry. Diesel pumps are not an optimal solution as these consume an exhaustible source of energy (diesel) at a fast rate. Additionally, emissions from these pumps are harmful and pumping capacity is also restricted.

Solar pumps overcome all these challenges. These use renewable energy as the source of power, which makes them environment-friendly while cutting down the complete cost of electricity or fuel.

Swapnil S. Ingole, country manager - India, Galcon, explains, "Think of an area where the water source is one kilometre away from the farmland and there is no electricity. The water has to be collected through pump pressure only. This is where solar pumps become very important."

S.S. Pradhan, general manager - sales, Rotomag Motors and Controls says, "In India, sunlight is available for around 305 days throughout a year. Water can be easily pumped on all these days and stored. Most of the remaining 60 are rainy days while the rest dry days can be easily managed with the help of the stored

Case study: How Dhundi village multiplied its income

In 2015, farm-owners at Dhundi village, Gujarat, installed solar pumps in their fields. The village lacked electricity, which initially drove them towards diesel pumps. They incurred expenses of ₹ 20,000-₹ 25,000 per year on fuel and maintenance of these diesel pumps. Additionally, to run the oil engine, they required skilled labour, including personnel for full-time system monitoring and emergency situations.

So the solar pump installation project was initiated by six farmers—three installing 5HP pumps and the rest going for 7.5HP. The farmers paid 20 per cent of the complete cost, while the rest (up to ₹ 4 million) was funded by International Water Management Institute (IWMI). The solar pumps installation completely cut down the expenses on diesel and other factors. The farm owners reported absolutely no operational and maintenance costs with solar pumps.

Later, considering the fact that farmers irrigate their land for fewer than 100 days a year, multiple farmers owning solar pumps formed a microgrid, which would create excess energy that they could sell. The process allowed them to earn ₹ 7.13 per kilowatt-hour of electricity generated. Even during short days of winters, farmers can generate 35-45 units of electricity and earn up to ₹ 8000-₹ 9000 per month. Additionally, they are able to sell excess water pumped from their tube wells and earn up to ₹ 250 for four hours of pumping every day.

water. Therefore a solar pump covers almost the whole year's water supply."

More importantly, solar pumps can substantially multiply farm-owners' income. Pradhan explains, "Using conventional pumps, farm owners could produce a volume which would earn them barely ₹ 100,000-₹ 200,000 a year. With solar pumps, they can produce throughout the year and now their average income has become ₹ 300,000-₹ 400,000 annually. So, essentially, solar pumps can increase their incomes by up to three times."

Investment, subsidies and payback

Agricultural solar pumps are powerful machines with high specifications. Ingole says, "The price of agricultural solar pumps starts from ₹ 50,000 per HP and goes up to ₹ 350,000 per HP. It depends on the features of the pump. For instance, basic pumps will require their panels to be positioned manually based on the direction of the sunlight. The costlier ones have automated controls for the same. Another price variation depends on the power consumption specification in HP. The price per HP is discounted for higher-powered pumps. For example, if a 1HP pump costs ₹ 100,000, the rate may be discounted to ₹ 90,000 per HP for a 5HP pump. Additional cost will include transportation, installation and other service charges."

The government provides subsidies to make solar pumps available to the agri-community throughout the nation.

Pradhan informs, "The Ministry of New and Renewable Energy (MNRE) has dedicated a Central Financial

Assistance (CFA) of 25 per cent, while the states are providing 50-70 per cent of the subsidy. So farm owners have to pay barely 10-20 per cent of the total pump cost. Pumps are manufactured with a five-year warranty. Therefore a chunk of the payback period is already covered and ensured."

Talking about the return on investment, Ingole shares, "According to market experts, the payback period of solar pumps is approximately 10 years. The complete saving on different operational expenses will bring substantial monetary returns even after the payback is achieved."

Pradhan feels that "Compared to diesel or electric pumps, solar pumps can achieve an ROI within 6-10 years for the investor." He highlights, "For the government as an investor, the scope of payback is even larger. Right now, we have a national electricity generation capacity of 350,000MW. Power requirement will reach 650,000MW by 2022. The government is incurring an expense of ₹ 800-₹ 900 billion by providing electricity generated from coal and other exhaustive resources to rural areas for free. If this power is instead generated using solar *in situ*, by installing solar pumps, the government can make major savings on resources."

The final word

Currently, there are not many solar DC pump manufacturers in India. Once the manufacturing capacity expands, pump costs will come down. While majority of the installations are being done through government projects, a decreasing upfront cost will attract more private investors. **EFY**

—Paramik Chakraborty, technical journalist, EFY